

A curvature diagnostic for continuous-RSB criticality versus kinematic volume-divergence in disordered systems

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Abstract

A diverging response is ambiguous: it can mark genuine interacting criticality, or a purely *kinematic* explosion of a metric *volume* while nothing critical happens. We give a sharp information-geometric criterion that separates them, built on the Ruppeiner scalar curvature R : a genuine transition is a diverging metric *component* on a non-degenerate, *positive-definite* metric, whereas a $\det g \rightarrow 0$ or coordinate-degenerate divergence is spurious and a manifold lacking the order parameter’s conjugate field is blind. “ $|R| \rightarrow \infty$ ” alone proves nothing. We validate the criterion on three exactly-solvable archetypes from distinct venues. (i) *Learning*: ridge regression at the interpolation threshold—double descent is kinematic, $\det g \sim \lambda^{-3/2}$ while R stays flat-to-bounded. (ii) *Spin glass*: Sherrington–Kirkpatrick approached along the de Almeida–Thouless line, where $|R| \sim \chi_{\text{SG}}^2$ diverges on a positive-definite metric. (iii) *Storage/jamming*: the spherical perceptron, which carries *both* sides in one model—convex storage is kinematic, while non-convex (negative-margin) storage is a continuous replicon at which $|R| \sim \chi^2$ diverges on a positive-definite finite-temperature metric, a signature the natural (structural) coordinate fails to reveal. The phases are classical; the contribution is their uniform curvature *classification* across venues—learning, spin glasses, and constraint satisfaction—of one subject: continuous-RSB criticality versus kinematic volume-divergence in disordered systems. We further establish a topological trichotomy for the genuine signature and argue that for connected, over-parametrized network posteriors it is vacuous—routing such architectures from the curvature diagnostic to the local learning coefficient.

1 Introduction

The Ruppeiner scalar curvature R of a thermodynamic state space is a geometric measure of fluctuations, diverging at ordinary critical points where the correlation volume becomes macroscopic [1, 2]. One-dimensional curvature claims are vacuous; on a two-parameter manifold R is a non-trivial invariant.

A recurring confusion motivates this paper. Modern high-dimensional inference exhibits divergences—most famously the double-descent peak of the generalization error at the interpolation threshold [3]—that *look* like phase transitions but need not be. The question “is this divergence a genuine transition?” has, we argue, a clean geometric answer, and the answer is not “does something blow up.” A metric *volume* $\det g$ can diverge for purely kinematic reasons while the intrinsic curvature stays bounded; conversely, $|R| \rightarrow \infty$ on a degenerate metric can be a coordinate artifact. The diagnostic must therefore be *refined*.

We make this precise (Sec. 2) and validate it on three exactly-solvable archetypes (Secs. 3–5). The archetypes are deliberately drawn from three different *venues*—supervised learning, mean-field spin glasses, and constraint satisfaction/jamming—because our central claim is that the dichotomy is not about any one of them. “The geometry of learning transitions” would name the venue, not the subject. The subject is the dichotomy between *continuous-RSB criticality* and *kinematic volume-divergence in disordered systems*.

Status of the claims. We follow a strict claim hygiene. All three archetypes are verified ([V]). For the perceptron (Sec. 5) we keep separate two statements that are easily conflated: the divergence of the *overlap susceptibility* (Sec. 5.2) and the literal *curvature* statement $|R| \rightarrow \infty$ (Sec. 5.3). The latter is the more demanding because “ $|R| \sim$ correlation volume” is Riemannian: it is established here on the finite-temperature (β, ε) manifold, where both coordinates are conjugate fields and the metric is positive-definite by construction. A first computation tuned by the structural load α gives the same divergence on an *indefinite* metric and is recorded as the cautionary intermediate it is.

2 The refined diagnostic

Let $\psi(\theta^1, \theta^2) = \log Z$ be the log-partition function of a two-parameter exponential family in natural parameters $\theta = (\theta^1, \theta^2)$, with sufficient statistics $T = (T_1, T_2)$. The Fisher information / thermodynamic metric is the Hessian

$$g_{ij} = \partial_{\theta^i} \partial_{\theta^j} \psi = \text{Cov}(T_i, T_j), \quad (1)$$

positive-semidefinite by construction. The Ruppeiner scalar curvature of (1) is, in the program’s sphere-positive convention,

$$R = -\frac{N}{2(\det g)^2}, \quad (2)$$

with N the standard determinant of second and third derivatives of ψ (Appendix; the engine is pinned to $R = -1$ on the normal family). The sign of R is convention-dependent; only the *invariant* distinction between $|R|$ bounded and $|R| \rightarrow \infty$ carries physical content. Equation (2) already exposes the logical engine of the diagnostic: a *diverging* $\det g$ *suppresses* $|R|$ (the kinematic mode—a volume that explodes faster than the numerator), whereas a *vanishing* $\det g$ *inflates* it (the spurious mode); only a diverging numerator component on a bounded-below, positive $\det g$ produces a genuine divergence.

Definition 1 (Refined criticality diagnostic). A divergence is a *genuine continuous transition* iff $|R| \rightarrow \infty$ is produced by a diverging metric *component* $g_{\varepsilon\varepsilon} \rightarrow \infty$ on a metric that remains *non-degenerate and positive-definite* ($\det g > 0$, well-conditioned), where ε is the natural field conjugate to the order parameter. Three configurations fail the criterion — the first two are disqualified divergences, the third sees no transition at all:

- *Kinematic (volume)*. $\det g \rightarrow \infty$ while $|R|$ stays bounded: the metric volume explodes, the intrinsic curvature does not. (Double descent, Sec. 3.)
- *Spurious (degenerate)*. $|R| \rightarrow \infty$ driven by $\det g \rightarrow 0$ or by functionally dependent coordinates (e.g. $E \propto M^2$, Curie–Weiss): a coordinate artifact, not a fluctuation.
- *Blind*. A manifold lacking the order parameter’s conjugate field as a coordinate sees no transition at all.

The discipline of Definition 1 is that “ $|R| \rightarrow \infty$ ” alone proves nothing; the mechanism and the signature must both be checked.

3 Kinematic archetype: ridge regression and double descent

Consider the Gibbs measure over student weights of a linear (Gaussian) teacher–student problem, $P(w) \propto \exp(-\beta E - \beta\lambda \frac{1}{2}\|w\|^2)$ with $E = \frac{1}{2}\|y - Xw/\sqrt{N}\|^2$ and a (possibly noisy) teacher $y = Xw_\star/\sqrt{N} + \sigma\xi$. The sufficient statistics are $(E, \frac{1}{2}\|w\|^2)$ and the natural parameters $(\theta^1, \theta^2) = (-\beta, -\beta\lambda)$ —the faithful analog of a gas’s $(-\beta, \log z)$, with the ridge field λ in the fugacity role.

3.1 The exact manifold

Because E is quadratic in w , $\log Z$ is a Gaussian integral, diagonal in the eigenbasis $\{v_i\}$ of the sample covariance $M = X^\top X/N$ with eigenvalues $\{\mu_i\}$ and teacher projections $d_i = \langle v_i, X^\top y/\sqrt{N} \rangle$:

$$\log Z(\theta) = \text{const} - \frac{1}{2} \sum_i \log k_i + \frac{\beta^2}{2} \sum_i \frac{d_i^2}{k_i}, \quad k_i = \beta(\mu_i + \lambda). \quad (3)$$

The metric (1) and all third derivatives are therefore closed-form spectral sums; the fluctuation that controls the transition is the variance term $g \supset \sum_i k_i^{-2}$. The interpolation threshold $\alpha = P/N = 1$ is the candidate transition: there the Marchenko–Pastur (MP) soft edge touches zero, so the spectral density behaves as $\rho(\mu) \sim (\pi\sqrt{\mu})^{-1}$ as $\mu \rightarrow 0$, and as the ridge $\lambda \rightarrow 0$ the softest mode $k_{\min} = \beta(\mu_{\min} + \lambda) \rightarrow 0$.

3.2 The kinematic verdict

At $\beta = 1, \alpha = 1$ the limit $\lambda \rightarrow 0$ produces a textbook-looking divergence that the diagnostic classifies as *kinematic* (receipt `module L_ridge_curvature.py`; analytic Hessian validated against finite differences of $\log Z$ to max. relative error 8.9×10^{-6} , engine pinned to $R = -1$ on the normal family):

quantity	as $\lambda \rightarrow 0$	reading
metric volume $\det g$	diverges $\sim \lambda^{-3/2}$	MP soft-edge fluctuation explosion
metric rank	collapses to rank 1 (defect $\rightarrow 1.000$)	one soft mode dominates
R , noiseless teacher	$\rightarrow 0, \sim \lambda$ ($-2.0\text{e-}5 \rightarrow -8.1\text{e-}10$)	asymptotically flat
R , noisy teacher ($\sigma = 0.5$)	bounded $\approx -1 \times 10^{-4}$, seed-robust	bounded floor, never diverges

The diverging volume is a direct soft-edge effect: from (3) the singular metric component is $\sum_i k_i^{-2} \rightarrow N \int_0 \rho(\mu) [\beta(\mu + \lambda)]^{-2} d\mu \sim \int_0 \mu^{-1/2} (\mu + \lambda)^{-2} d\mu \propto \lambda^{-3/2}$, so $\det g \sim \lambda^{-3/2}$ while the metric collapses onto the single softest mode (the defect $\rightarrow 1$).

3.3 Mechanism: a free versus a sourced soft mode

The divergence is thus *entirely* a metric-volume effect: $\det g \rightarrow \infty$ while $|R|$ is bounded (noisy) or vanishing (noiseless). It is the exact cancellation of singular powers in $R = -N/(2(\det g)^2)$, with N and $(\det g)^2$ diverging at *matching* rates, so that R stays finite whenever the diverging volume is sourced by a free log-determinant sector—the same flat-sector cancellation that keeps thermodynamic curvature finite at a limiting-temperature transition in product-form gases. The mechanism is transparent at the soft mode: with a noiseless teacher the soft direction is a *free* log-determinant mode (signal weight $d_{\min}^2 \sim \mu_{\min}^2 \rightarrow 0$ vanishes with the eigenvalue), giving exactly flat curvature, the classical ideal-gas baseline; noise gives that mode a *source* ($d_{\min}^2 \rightarrow$ the noise floor $\sim \sigma^2$), lifting R to a bounded nonzero value. Neither diverges. Forty-digit arithmetic is required, not optional: the rank-1 collapse costs ~ 11 digits to cancellation, and at single precision the bounded R masquerades as a divergence—a standing caution for numerical thermodynamic geometry in high dimension.

4 Genuine archetype: the SK spin glass at the dAT line

The Sherrington–Kirkpatrick model [4] is the positive control: a genuine interacting transition, reached without any replica-symmetry-breaking ansatz.

4.1 The two-replica construction

We stay replica-symmetric and *approach* (never cross) the de Almeida–Thouless (dAT) instability [5]. Take two real replicas s^1, s^2 and couple them by a field ε conjugate to their overlap $O = \sum_i s_i^1 s_i^2$; the disorder-averaged two-replica RS free energy $\psi_2(\beta, h, \varepsilon)$ is then a function on a manifold whose ε -direction *is* the order-parameter channel. At $\varepsilon = 0$ the replicas decouple, $\psi_2 = 2\psi_1$ (verified to $|\text{diff}| = 0$), so ε is a clean perturbation of the single-replica state and the ε -metric component is the spin-glass susceptibility $g_{\varepsilon\varepsilon} = \beta \partial p / \partial \varepsilon = \beta \chi_{\text{SG}}$, $p = \langle O \rangle / N$ the inter-replica overlap.

4.2 The $h = 0$ closed form

At $h = 0$ the construction is fully explicit:

$$\psi_2(\beta, \varepsilon) = \frac{\beta^2}{2}(1 - p^2) + \log 4 + \log \cosh(\beta^2 p + \beta \varepsilon), \quad p = \tanh(\beta^2 p + \beta \varepsilon). \quad (4)$$

For $\beta < 1$ the saddle is $p = 0$ at $\varepsilon = 0$; differentiating the self-consistency, $\partial p / \partial \varepsilon|_0 = \beta \text{sech}^2(0) / (1 - \beta^2 \text{sech}^2(0)) = \beta / (1 - \beta^2)$, so with the replicon $\lambda_{\text{repl}} \equiv 1 - \beta^2$,

$$g_{\varepsilon\varepsilon} = \frac{\beta^2}{\lambda_{\text{repl}}}, \quad R_{\text{bare}} = \frac{4}{\beta^2} \text{ (bounded: bare manifold } \textit{blind}), \quad R_\varepsilon = \frac{2}{\beta^2 \lambda_{\text{repl}}^2} \xrightarrow{\lambda_{\text{repl}} \rightarrow 0} \infty, \quad \det g = \frac{\beta^2}{\lambda_{\text{repl}}} \quad (5)$$

So $|R_\varepsilon| \sim \chi_{\text{SG}}^2$ diverges on a *strictly positive-definite* metric—a genuine transition by Definition 1 (e.g. $R_\varepsilon = 68, 233, 1328$ at $\beta = 0.90, 0.95, 0.98$), the exact prototype the perceptron reproduces in Sec. 5.3. The bare (β, h) manifold, lacking the ε -coordinate, sees only the bounded $R_{\text{bare}} = 4/\beta^2$: it is *blind*.

4.3 Along the de Almeida–Thouless line

Off the symmetric axis there is no closed form: the coupled (p, q) saddle is solved on Gauss–Hermite nodes and $g_{\varepsilon\varepsilon}$ obtained by *implicit differentiation* (linear response). It scales cleanly as $g_{\varepsilon\varepsilon} = \beta \chi_{\text{SG}} \sim C(h) / \lambda_{\text{AT}}$ all along the dAT line, the constant $C(h)$ stabilizing as the replicon $\lambda_{\text{AT}} \rightarrow 0$:

h	λ_{AT}	β	$g_{\varepsilon\varepsilon}$ (implicit)	$g_{\varepsilon\varepsilon}$ (fin. diff.)	$\chi_{\text{SG}} \lambda_{\text{AT}}$
0.3	0.30	1.013	3.83	3.83	1.13
0.3	0.10	1.342	17.36	17.98	1.29
0.3	0.02	1.495	97.46	30.98	1.30
0.5	0.30	1.265	4.35	4.35	1.03
0.5	0.10	1.662	17.76	18.29	1.07
0.5	0.02	1.827	97.79	32.03	1.07

The product $\chi_{\text{SG}} \lambda_{\text{AT}} \rightarrow C(h)$ flattens to three digits, confirming $g_{\varepsilon\varepsilon} \sim 1/\lambda_{\text{AT}} \rightarrow \infty$; note the finite-difference column *drifts* below the stable implicit value as $\lambda_{\text{AT}} \rightarrow 0$ (30.98 vs. 97.46 at $\lambda_{\text{AT}} = 0.02$), the breakdown quantified in Remark 1. The divergence mechanism is $\det(I - M) \rightarrow 0$ —the two-replica saddle’s stability matrix going singular, which *is* the replicon condition—with $\det g > 0$ and growing throughout, while the bare (β, h) curvature stays bounded (≈ 16 at $h = 0.3$): blindness confirmed off-axis. The diagnostic tracks the replicon wherever it sits in the (T, h) plane [21, receipt `module_L_SK_converse.py`].

Remark 1 (Numerical caveat, honestly recorded). Third-derivative finite differences of R break down for $\lambda_{\text{AT}} \lesssim 0.05$ (in double precision on an `fsolve`’d saddle, R sign-flips and the finite-difference $g_{\varepsilon\varepsilon}$ itself drifts, 31 vs. 97 at $\lambda_{\text{AT}} = 0.02$). The $|R| \rightarrow \infty$ claim therefore rests on

the $h = 0$ closed form (5), the stable linear-response $g_{\varepsilon\varepsilon}$ (clean to $\lambda_{AT} = 0.02$), and the R growth where the third difference is still reliable ($\lambda_{AT} \gtrsim 0.1$: $26 \rightarrow 41 \rightarrow 99$)—never on the edge values. The same caveat governs the perceptron curvature (Sec. 5.3), which is likewise built from a soft-mode/linear-response construction rather than naive edge differences.

5 The third archetype: the spherical perceptron carries both sides

The Gardner storage problem for a perceptron [6, 7] is, geometrically, a single model that realises *both* sides of the dichotomy as the weight geometry is varied. For weights w ($\|w\|^2 = N$ spherical, or $w \in \{\pm 1\}^N$ Ising), $p = \alpha N$ random patterns at margin κ , the Gardner volume $V = \int d\mu(w) \prod_{\mu} \Theta(\sigma^{\mu}(w\xi^{\mu})/\sqrt{N} - \kappa)$ has the replica-symmetric free entropy $s(q) = G_S(q) + \alpha G_E(q)$ with the energetic term

$$G_E(q) = \int D t \log H(u), \quad u = \frac{\kappa - \sqrt{q} t}{\sqrt{1-q}}, \quad H(x) = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2}), \quad (6)$$

($D t$ the standard Gaussian measure) and the spherical entropic term $G_S(q) = \frac{1}{2}[\log(1-q) + q/(1-q)]$. We emphasise that (6) is the log-volume of a *constraint-satisfaction* problem; the transition below is jamming, not generalization.

The *phases* of this model are classical: the Gardner capacity [6, 7], the frozen-1RSB capacity of the Ising case [8], and the continuous-RSB / jamming instability of the negative-margin case are all established results. We claim none of them as new. The contribution of this section is not the phase diagram but its *classification by curvature signature*: we show that the *same* information-geometric criterion (Definition 1) that separates double descent from SK assigns each of these known phases to the correct tier, and in particular that one model supplies both a kinematic and a genuine entry. The lens is the object of study; the phases validate it.

5.1 Convex storage is kinematic; Ising storage is first-order

For $\kappa \geq 0$ the problem is convex: the RS solution is exact up to capacity and there is no replicon instability. The only transition is $q \rightarrow 1$ at the Gardner capacity $\alpha_c(\kappa) = [\int_{-\kappa}^{\infty} D t (t + \kappa)^2]^{-1}$ (so $\alpha_c(0) = 2$ exactly, verified), a *metric-volume* effect—the storage twin of double descent, the kinematic side. For Ising weights the storage transition is instead *frozen-1RSB*: the RS free entropy starts at $s(0) = \log 2$ and goes negative, crossing zero at $\alpha_{RS} = 0.83308$, coinciding with the Krauth–Mézard capacity 0.833 [8]; a negative quenched entropy for discrete weights is impossible, so the solution space freezes discontinuously (q jumps) and the continuous-replicon route does not apply. (The Ising *learning* transition is likewise first order [9].)

5.2 Non-convex storage: susceptibility-genuine $[V]$

For $\kappa < 0$ the problem is non-convex and a continuous replicon appears below capacity. From the RS saddle $q/(1-q) = \alpha \int D t (H'/H)^2$ the replicon (AT) eigenvalue is

$$\lambda_{\text{repl}} = 1 - \alpha \int D t [G'_1(u)]^2, \quad G_1 = H'/H, \quad G'_1 = -u G_1 - G_1^2. \quad (7)$$

Equation (7) is anchored to Gardner: at $\kappa = 0$ it vanishes precisely at $\alpha = 2 = \alpha_c$ (the AT line meets capacity, marginal). For $\kappa < 0$ it crosses zero *continuously* at $\alpha_{AT}(\kappa) < \alpha_c(\kappa)$ (e.g. $\alpha_{AT} = 2.32, 2.67, 3.05, 3.95$ for $\kappa = -0.1, -0.2, -0.3, -0.5$), and the overlap susceptibility $\chi = 1/\lambda_{\text{repl}}$ diverges continuously, $\chi \cdot (\alpha_{AT} - \alpha) \rightarrow \text{const}$ (≈ 3.22 at $\kappa = -0.5$)—a clean $1/(\alpha_{AT} - \alpha)$ power law, the direct twin of SK’s $\chi_{\text{SG}} \sim 1/\lambda_{AT}$. Since ε (conjugate to the overlap) is a genuine

natural field, $g_{\varepsilon\varepsilon} \sim 1/\lambda_{\text{repl}}$ is a true conjugate-field susceptibility whose divergence is coordinate-independent, Gardner-anchored, and component-driven. **By the susceptibility criterion the non-convex spherical perceptron sits on the genuine side; this is verified [V] [21, receipt module_L.perceptron_replicon.py].**

5.3 The literal curvature: [V], on a positive-definite manifold

On the RS side the overlap sector is a Gaussian soft mode, so the ε -augmented free energy is $\psi(c, \varepsilon) = 2\psi_0(c) + \chi(c)\varepsilon^2/2 + O(\varepsilon^4)$ in a *generic* tuning coordinate c , with $\chi = 1/\lambda_{\text{repl}}$ the overlap susceptibility. We instantiate c two ways below—first the structural load α , which fails the signature test, then the natural field β , which passes—to show the diagnostic is doing real work.

At $\varepsilon = 0$ the engine inputs are $g_{\varepsilon\varepsilon} = \chi$ (singular), $g_{cc} = 2\psi_0''$ and $g_{c\varepsilon} = 0$ (regular), and among the third derivatives only the cross-term $\partial_c g_{\varepsilon\varepsilon} = \chi' \sim \chi^2$ is singular (since c tunes λ_{repl}), while $\partial_\varepsilon^3 \psi = 0$ and the rest are $O(1)$. The numerator N of (2) is then dominated by the single term $\propto (\partial_c g_{\varepsilon\varepsilon})^2 \sim \chi^4$, whereas $\det g \sim g_{cc} \chi \sim \chi$; hence $R = -N/(2(\det g)^2) \sim \chi^4/\chi^2 = \chi^2$, i.e. $|R| \sim \chi^2 \sim 1/(c_{AT} - c)^2$, the curvature diverging as the *square* of the susceptibility, exactly as SK’s $R_\varepsilon \sim \chi_{\text{SG}}^2$. Crucially this is a genuine divergence, *not* the ridge cancellation of Sec. 3: there the singular powers of N and $(\det g)^2$ matched and cancelled, whereas here the singular cross-derivative $\partial_c g_{\varepsilon\varepsilon}$ survives uncanceled. Freezing χ (i.e. setting $\chi' = 0$) sends $R \rightarrow 0$, confirming the divergence is a *component* effect, not a background one.

The signature, however, depends on *which* tuning coordinate is used, and this is where Definition 1 earns its keep. Tuning with the load α (a *structural* parameter, not a natural field) gives $g_{\alpha\alpha} = 2s'' < 0$ (the free entropy is concave): $|R| \rightarrow \infty$ but on an *indefinite* metric ($\det g < 0$), which does not establish the Riemannian statement “ $|R| \sim$ correlation volume.” We therefore tune instead with a genuine natural field. At finite temperature the storage problem is a Gibbs measure with inverse temperature β conjugate to the constraint energy; the soft constraint $a = e^{-\beta}$ gives $m = a + (1 - a)H(u)$ and $M_1 = (1 - a)H'/m$, with the replicon retaining its form, $\lambda_{\text{repl}} = 1 - \alpha \int D t [M_1']^2$, $M_1' = -uM_1 - M_1^2$ (recovering (7) as $\beta \rightarrow \infty$). Now β is a natural field, so $g_{\beta\beta} = 2\psi_0'' = 2\text{Var}(\text{energy})/N > 0$, *and* it tunes the replicon, $\lambda_{\text{repl}}(\beta) \rightarrow 0$ at a finite β_{AT} . On the (β, ε) manifold both coordinates are conjugate fields, so the metric is a genuine covariance, positive-definite by construction. The receipt confirms it at *two* operating points in the finite- T RSB window:

$$\det g > 0 \text{ throughout and diverging,} \quad |R| \sim \chi^2 \sim \frac{1}{(\beta_{AT} - \beta)^2}, \quad |R| \cdot (\beta_{AT} - \beta)^2 \rightarrow \begin{cases} 11.8 & (\kappa = -0.5, \alpha = 0.5) \\ 12.9 & (\kappa = -0.3, \alpha = 0.3) \end{cases} \quad (8)$$

so the χ^2 law is robust across the line, not a one-point fit. This is a diverging metric *component* ($g_{\varepsilon\varepsilon} = \chi \rightarrow \infty$) on a *non-degenerate, positive-definite* metric—the genuine side of Definition 1 in full, the literal Riemannian twin of (5). **Curvature-genuine for the perceptron storage/jamming archetype is verified [V] [21, receipt module_L.perceptron_finiteT.py; the indefinite (α, ε) first pass is module_L.perceptron_curvature.py].**

6 The genuine gate and the topological trichotomy

The three archetypes of Secs. 3–5 are *non-singular*: their Fisher metrics are non-degenerate away from the transition, and the diagnostic classifies them cleanly. Real overparametrized networks are the *singular* case — non-identifiable, with positive-dimensional degeneracies on which $\det I(w) \rightarrow 0$ — so whether the genuine-side construction survives there is a separate question, decided on paper before any sampler (the analysis is archived with the receipts [21]). This section states the gate, refutes an unconditional closure, and reduces the genuine side to a structural trichotomy.

6.1 From kinematic to genuine

Section 3 established that double descent is kinematic: a covariance degeneracy under which the scalar curvature stays bounded despite a diverging volume. We ask the converse — under what conditions is a curvature divergence *genuine*, a signature of interacting criticality rather than a coordinate or volume artifact? For the solvable archetypes the criterion is sharp (Definition 1): genuine criticality requires a diverging metric *component* on a non-degenerate, positive-definite thermodynamic metric. Closing the genuine side for a *singular* posterior therefore means showing the singular structure forces $\det g \rightarrow 0$ at the transition.

6.2 Refutation of unconditional closure

It does not, in general. A singular minimum has density of states $\rho(E) \sim E^{\lambda-1}$, with λ the RLCT — the same exponent that governs the Bayes free energy $F_n \approx \lambda \log n$. Then $Z(\beta) = \Gamma(\lambda) \beta^{-\lambda}$, so $\text{Var}(E) = \lambda/\beta^2$ and the energy-fluctuation component is suppressed, $g_{\beta\beta} \sim \lambda/\beta^2 \rightarrow 0$ — but only in the singular ($\beta \rightarrow \infty$) limit. The curvature is then governed by the competition

$$\det g = g_{\beta\beta} g_{\varepsilon\varepsilon} - g_{\beta\varepsilon}^2 \sim \frac{\lambda}{\beta^2} \chi - g_{\beta\varepsilon}^2, \quad (9)$$

between this singular suppression and the overlap susceptibility $\chi = g_{\varepsilon\varepsilon}$ that drives genuine criticality at the de Almeida–Thouless line. At a *finite*- β transition $g_{\beta\beta} = O(1)$ while $\chi \rightarrow \infty$, so $\det g \rightarrow +\infty$ and the genuine signature survives; the suppression, a $\beta \rightarrow \infty$ effect, never reaches it. The closure can bite only where the criticality *co-locates* with the singular limit, $\chi \sim \beta^p$: then $\det g \sim \lambda \beta^{p-2} - g_{\beta\varepsilon}^2$, and the threshold is $p = 2$ — $p < 2$ closes the gate ($\det g \rightarrow 0$, F-spurious), $p > 2$ leaves it open, $p = 2$ is borderline. Here p is the *susceptibility exponent* in β , not the order of any p -spin interaction. (This corrects a first pass that claimed unconditional closure; the finite- β case refutes it.)

6.3 The topological trichotomy

Because $\chi = N \text{Var}(q)$ diverges *only* at a continuous-RSB transition, the genuine side is decided not by a rate but by *where* that transition sits relative to the singular limit — a structural question with exactly three branches (Table 1).

branch	RSB structure	$\det g$ as critical	verdict
1	finite- β RSB ($\beta_{AT} < \infty$)	$\rightarrow +\infty$ ($g_{\beta\beta} = O(1)$, $\chi \rightarrow \infty$)	genuine survives; gate OPEN
2	zero- T RSB ($\beta \rightarrow \infty$), $\chi \sim \beta^p$	$\sim \lambda \beta^{p-2} - g_{\beta\varepsilon}^2$	$p < 2$ CLOSED / $p > 2$ OPEN / $p = 2$ BORDERLINE
3	no RSB (connected manifold)	— (χ saturates, finite)	vacuous; closed for <i>absence</i> of criticality

Table 1: The genuine-side trichotomy for a singular posterior. Branch 1 is realized by the spherical perceptron (Sec. 5) and SK at the dAT line (Sec. 4) — which is exactly why their genuine results are $[V]$ and a control was even conceivable. Branch 2 has no exactly-solvable archetype here, and the model for $\chi(\beta)$ that would fix p is left open. Branch 3 is the mode-connected case of §6.4.

In Branch 1 the singular suppression is irrelevant — it lives at $\beta \rightarrow \infty$, the transition is at finite β — so the genuine signature is intact, at the cost that an equilibrium sampler is then needed after all. Branch 2 is the only branch where the (9) race is decisive. In Branch 3 two replicas spread over a connected manifold, so χ saturates to a finite, β -independent constant and never diverges: the genuine signal is vacuous, the gate closed not by suppression but by the *absence* of a transition to detect. Whether a real network’s posterior is Branch 1, 2, or 3 is the spin-glass-of-nets question, addressed empirically next.

6.4 The handoff (conditional)

Empirically and, for wide networks, provably, trained-network solution sets are connected — by low-loss curves [12, 13], as connected sublevel sets under over-parametrization [14], and linearly once permutation symmetry is quotiented [15, 16]. Connectivity is the geometric signature of absent ergodicity breaking, placing such posteriors in Branch 3, where the overlap susceptibility does not diverge and the genuine Ruppeiner signature is vacuous. The spin-glass picture of network landscapes [17] is an analogy under strong assumptions; the most direct dynamical comparison finds true glassiness only in the under-parametrized regime [18]. We therefore *argue* — not prove — that for over-parametrized networks the genuine-criticality construction is vacuous; whether any real posterior instead realizes a finite- β (Branch 1) or zero- T (Branch 2) transition remains open. Where the signature is vacuous, the appropriate invariant is the singularity order — the RLCT [19], estimated as the LLC [20]. Two distinct facts co-locate on one object: the connected singular set is where the Ruppeiner signature goes vacuous (no RSB) and, simultaneously, where the Fisher information degenerates ($\det I \rightarrow 0$); it is the latter that makes the LLC the natural measurement there. Two facts, one object, not one mechanism — which is why singular learning theory measures the LLC in exactly this regime.

6.5 An empirical probe: the branch placement, bounded by measurement

The handoff of §6.4 is argued. We close with a first *empirical* probe of the branch placement on a samplable surrogate — not a settlement, but a measured bound where the posterior is reachable.

Setup. A two-layer tanh network (width H , scalar output) on an under-determined teacher–student task ($n = 24 \ll$ parameters) — the regime where the question is non-trivial, since a realizable, identifiable task collapses to a unique solution and a vacuous $q \rightarrow 1$. The tempered posterior $\pi_\beta \propto \pi_0(w) e^{-\beta n L(w)}$ (prior π_0 as base measure, likelihood tempered, so $\beta \rightarrow 0$ is a proper melt) is sampled by Metropolis-adjusted Langevin with parallel tempering on a ladder spaced equally in thermodynamic length $\mathcal{L} = \int \sqrt{C(\beta)} d \ln \beta$, $C = \beta^2 \text{Var}(E)$; the rung count is thereby an *output*, not a guess. The order parameter is the *function-space* overlap q of two posterior replicas on a fixed holdout probe — permutation-invariant by construction, so weight-space symmetry cannot counterfeit it — with susceptibility $\chi_F = N \text{Var}(q)$.

Certification. A width is admitted only on evidence that the sampler sees the whole posterior, not a sub-region: (i) two replicas *warm-started from independently-trained solutions*, verified separated ($q_{\text{init}} < 0.9$), must converge to a common cold- q distribution — init-independence read as a direct connectivity test; (ii) tempering round-trip flow exceeding one per replica with the inter-init flow spread closed; (iii) function-space $\hat{R} < 1.1$. Widths failing any gate are recorded INCONCLUSIVE and not read.

Result. Across the reachable widths the susceptibility *grows*: $\chi_F = 0.54 (H=32) \rightarrow 0.99 (H=64)$, both certified (separated replicas converged, $dq \sim 10^{-3}$; flow spread ≤ 0.03). At $H=128$ the machinery *cannot certify* — round-trip flow stalls below one per replica even on the thermodynamic-length ladder — so $\chi_F \approx 2.2$ there is inadmissible. This is a *bound*, not a null: the budget at which $H=128$ stalls is a quantitative floor on the mixing barrier, the genuinely-glassy corner where the branch-2/branch-3 distinction would be decided. Two controls discipline the read. An earlier, *unseparated-init* certification of $H=128$ was *refuted* by the adversarial test (its flow was init-dependent), confirming that the basin door must be shut before any growth is read. And the energy-side specific heat stays *broad* at every width (no sharpening peak), so the overlap-side growth is *not* corroborated by a thermodynamic-transition signature.

What it bounds. The branch question is now bounded by measurement rather than argument: a certified, basin-verified χ_F growth where the posterior is reachable; provably *unreachable* at the width where branch 2 and branch 3 would separate; and *uncorroborated* by the specific heat. The reachable evidence is ambiguous between sub-critical growth without a transition and a transition that cannot be sampled — the absent specific-heat peak leans the former, mildly consistent with the mode-connectivity (branch 3) reading of §6.4, but as a lean, not a settlement. Status [E-hw]: one task and architecture family, two certified widths; the receipts and the adversarial-certification protocol are archived with this work [21]. A stronger glassy-regime sampler for $H \geq 128$ and a denser width sweep are the deliberate next passes, neither required for the bound stated here.

7 Discussion

The four-tier picture (Table 2) is uniform across venues. A divergence earns “genuine” only as a positive-definite component blow-up; the metric-volume explosion of double descent, the coordinate-degeneracy of Curie–Weiss, and the blindness of a manifold without the conjugate field are its three foils.

system (venue)	metric $\det g$	scalar R
ideal Bose gas (BEC)	non-degenerate, diverging comp.	$ R \rightarrow \infty$
ridge / double descent (learning)	$\rightarrow \infty$ (rank-1)	$\rightarrow 0^-$ (noiseless) / bdd < 0 (noisy)
Curie–Weiss, dependent coords	$\rightarrow 0$ / indefinite	spurious $ R \rightarrow \infty$
bare SK (β, h)	finite	bounded
ε -augmented SK at dAT (spin glass)	> 0 , diverging comp.	$ R \rightarrow \infty$
spherical perceptron, $\kappa \geq 0$ (storage)	regular	bounded
spherical perceptron, $\kappa < 0$ (jamming)	> 0 , diverging comp.	$ R \sim \chi^2$

Table 2: The diagnostic across venues. The perceptron uniquely supplies both a kinematic and a genuine entry.

The ideal Bose gas at condensation (Table 2, first row) is the canonical classical exemplar of the genuine signature [2], included as the literature anchor; the three disordered-system archetypes below it are this paper’s contribution.

The organising claim is that none of these systems is special: the dichotomy is a statement about disordered systems with a continuous replicon, and learning is merely the venue in which the question is most often asked. None of the underlying phase structure is new—it is the classical content of [6, 7, 8, 4, 5]. What is verified ($[V]$) is the per-system curvature behaviour of each archetype; their *uniformity under a single criterion*—that these facts collectively constitute a general diagnostic separating genuine from kinematic—is the extrapolation we propose, supported by but not reducible to the three exactly-solvable cases. The discipline of Definition 1 (positive-definite signature, not merely $|R| \rightarrow \infty$) is load-bearing rather than pedantic: the (α, ε) versus (β, ε) contrast of Sec. 5.3 is exactly a case where the wrong coordinate produces a divergence that *looks* genuine on an indefinite metric and is not.

Related work. The closest published prior art is the curvature analysis of L_2 -regularized deep networks by Ersoy and Wiesner [11], who explain the regularization-driven *onset of learning*—a first-order transition into the under-parametrized phase, with further transition points and hysteresis as data complexity grows, and a reinterpretation of “grokking”—through the scalar (Ricci) curvature of the *error landscape*, extending the depth-dependent transition orders (second order for a single hidden layer, first order for deeper nets) of Ziyin and Ueda [10]. The

overlap is real, but the geometric objects are distinct. (i) *The metric*. Theirs is the embedding geometry of a fixed loss surface (its first and second fundamental forms; Gauss–Kronecker and Ricci scalars); ours is the Ruppeiner thermodynamic metric (1), the Hessian of a disorder-averaged free energy—a *covariance of fluctuations*. A deterministic loss surface carries no fluctuation content, so it cannot make the “ $|R| \sim$ correlation volume” reading that is the center of our diagnostic. (ii) *The control field*. Theirs is the L_2 penalty strength (which they also denote β); ours is a genuine inverse temperature, with the conjugate coordinate ε the order parameter’s, not a regularizer’s. (iii) *Disorder*. Their construction is a single deterministic realization, with no quenched average and no replicon; our kinematic-versus-genuine distinction turns on the stability of the replica-symmetric saddle—the replicon / de Almeida–Thouless line—which their setting does not contain. (iv) *The taxonomy*. Their first-/second-order classification (after Ziyin–Ueda) concerns the *continuity* of the collapse; ours (Definition 1) concerns whether a curvature divergence is genuine interacting criticality or a *kinematic* volume divergence—an orthogonal axis, since a transition that is second-order in their sense may be either kinematic or genuine in ours, a distinction their single-realization metric cannot draw. (v) *Degeneracy*. Where they observe that the Ricci scalar is numerically more stable than the Gauss–Kronecker curvature, our claims rest on *establishing* a non-degenerate, positive-definite metric on the relevant submanifold, without which $|R| \rightarrow \infty$ is disqualified as a coordinate artifact. (vi) *Scope*. Their results are numerical, on small networks; ours are closed-form on exactly-solvable mean-field archetypes (SK, the spherical perceptron) with exact capacities and AT lines. The two are complementary: theirs measures the shape of a deterministic loss surface, ours the fluctuation geometry of a disordered free energy, and the genuine-versus-kinematic distinction is not available on their object.

Limitations. The diagnostic presumes one can identify and adjoin the order parameter’s conjugate field ε ; absent that, the manifold is blind by construction, so the method *classifies* known transitions rather than discovering order parameters. Our validation is on mean-field, replica-symmetric (or RS-approached) exactly-solvable archetypes on two-parameter manifolds; extension to finite-dimensional systems, to full-RSB structure, and to higher-dimensional manifolds is open.

A The curvature engine and receipts

All curvatures use $R = -N/(2(\det g)^2)$ with $N = g_{xx}(g_{xy}g_{yyy} - g_{xyy}^2) - g_{xy}(g_{xx}g_{yyy} - g_{xyy}g_{xy}) + g_{yy}(g_{xx}g_{xyy} - g_{xxy}^2)$, pinned to $R = -1$ on the normal family. Five computational receipts reproduce every numerical claim: `module_L_ridge_curvature.py` (Sec. 3), `module_L_SK_converse.py` (Sec. 4), `module_L_perceptron_replicon.py` (the susceptibility, Sec. 5.2), `module_L_perceptron_finiteT.py` (the positive-definite (β, ε) curvature, Sec. 5.3), and the indefinite-metric first pass `module_L_perceptron_curva` (the cautionary (α, ε) intermediate). The anchors $\alpha_c(0) = 2$ and $s(0) = \log 2$ are exact to machine precision, the (β, ε) replicon is anchored to Gardner ($\beta \rightarrow \infty$) and the engine to $R = -1$.

Reproducibility. All receipts are CPU-only Python (`numpy/scipy`; `mpmath` at 40 decimal places where cancellation requires it, e.g. the rank-1 ridge metric). Gaussian integrals use 200-node Gauss–Hermite quadrature; saddles are solved by `brentq/fsolve` to tolerance 10^{-13} ; finite differences for R are used only where Remark 1 certifies them, with the load-bearing components obtained by closed form or implicit-differentiation linear response. The receipts and the gate document `T1_moduleL_perceptron_gate` are archived with this work [21].

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